

CareConnect: Concept Prototype 2

Workflow Overview

Phase 1: Ride Planning and Booking

The user enters all known appointments for the coming month or year that require TixiTaxi transport. The agent stores these rides, creates a structured overview, and sends the booking request to TixiTaxi.

Phase 2: Ride Management and Change Handling

Confirmed rides are stored in the system and remain visible to the user at all times. If plans change, the user can see the affected rides in the system. Cancellation, however, must currently still be done manually by phone with TixiTaxi. After cancellation, the user or agent updates the system record.

Phase 3: Invoice and Reimbursement Preparation

At the end of the month, the agent collects TixiTaxi invoices from email, checks them against the stored ride plan, processes additional uploaded receipts, and calculates private mileage claims for self-driven trips.

Phase 4: Final Claim Assembly and Submission

The agent prepares the reimbursement form, combines all supporting documents, drafts the email to SVA, and presents the full package for user review. Only after explicit approval is the submission sent.

Detailed Workflow

Phase 1: Planning and Sending Rides to TixiTaxi

User action

The user enters all already known appointments that require TixiTaxi transport for the coming month or year. (addition, self driven rides can be entered as well and these will be processed by the mileage Agent)

Examples:

- therapy appointments
- care institution visits
- medical appointments
- recurring weekly rides

Commented [BT1]: Shall we make clear definitions for those appointments. I suppose there will be a drop-down menu to choose the kind of appointment (also necessary for SVA). Is there an information in the form (using the same structure/wording)

Commented [BT2R1]: Alternative logic: Choose of Doctor and it is automatically connected the type of appointment (more user-friendly)

Commented [BT3R1]: Recurring Weekly Rides is just a time-relevant topic but not a type of appointment (mixed logic)

Agent tasks

The agent:

- stores all entered rides in the system
- structures them into a clear trip overview
- groups them by date, time, destination, and recurrence
- prepares a booking message for TixiTaxi
- sends the ride request to TixiTaxi

TixiTaxi confirmation logic

TixiTaxi then confirms the requested rides.

The system should apply the following rule:

- If the confirmed time differs by **30 minutes or less**, the ride is **automatically accepted**
- If the deviation is **more than 30 minutes**, the agent still stores the change, but the email to TixiTaxi should explicitly request that they **contact the user by phone** regarding the affected ride

Output of this phase

At the end of this phase, the system has:

- a full list of requested rides
- confirmation status for each ride
- accepted time adjustments where applicable
- a clean schedule of planned transport activities

Phase 2: Ride Visibility and Manual Cancellations

Once rides are confirmed, they are stored in the system as the official monthly ride plan.

User capabilities

The user can always:

- view all planned rides
- check ride status
- identify upcoming rides
- see which rides were confirmed or adjusted

Change handling

If plans change, the system should support visibility and documentation, but not full automation yet.

Commented [BT4]: Where is the grouping? Is it for the form that needs to be extracted or do you want to have it for the overview in the app or do you mean a search function?

For search function: Date, Destination, Recurrence (makes sense), for time I do not see a use case

For in-app overview: I think it only makes sense to have a overview structured by date (next appointment) with a status (completed or needs action)

Important limitation

If a ride must be cancelled, the user must currently:

- call TixiTaxi manually
- cancel the ride by phone
- remove or mark the ride as cancelled in the system

This is important because the workflow remains realistic and aligned with the current service reality instead of pretending full automation where it does not yet exist.

Output of this phase

The system maintains an updated and trustworthy ride record that reflects:

- planned rides
- completed rides
- cancelled rides
- adjusted rides

Phase 3: Invoice Matching, Receipt Handling, and Mileage Calculation

This phase happens at the end of the month and is split into three parallel sub-processes.

3.1 TixiTaxi Invoice Check from Email

At month-end, the user receives the TixiTaxi invoice by email.

Agent tasks

The agent:

- monitors the user's email inbox for TixiTaxi invoices
- identifies and extracts the invoice
- compares the invoice against the rides stored in the system
- checks whether billed rides match recorded rides
- flags mismatches such as:
 - missing ride
 - extra billed ride
 - incorrect date
 - incorrect number of rides

Goal

This ensures that the invoice is consistent with the planned and completed rides in the system before reimbursement is prepared.

3.2 Manual Upload of Other Receipts

Not all reimbursable expenses come through email. Some receipts must be uploaded manually by the user.

Examples:

- meal receipts
- SBB tickets
- other relevant receipts

Agent tasks

The agent:

- accepts manual receipt uploads
- stores the files in the correct month and category
- extracts relevant data from the receipt, such as:
 - date
 - amount
 - provider
 - category
- prepares the information for the SVA reimbursement form

Goal

The user no longer has to manually copy receipt data into the reimbursement file.

3.3 Mileage Agent for Self-Driven Trips

If parents drive the child themselves, these trips should also be claimable.

User input

The user enters:

- date and time
- start location
- destination

Agent tasks

The mileage agent:

- calculates the trip distance automatically
- determines the eligible reimbursement amount

- stores the mileage entry in the monthly reimbursement records

Goal

Private transport becomes just as easy to document as TixiTaxi rides.

4. Monthly Reimbursement Package Creation

At the end of the month, the system creates the final reimbursement package.

Agent tasks

The agent:

- gathers all TixiTaxi invoices
- gathers all manually uploaded receipts
- gathers all mileage records
- fills out the SVA reimbursement form automatically
- inserts the relevant expense information into the form
- merges supporting receipts into one PDF where needed
- attaches TixiTaxi invoices directly
- prepares a complete submission email to SVA

Attachment logic

- **TixiTaxi invoices** are attached directly
 - **Meal and other receipts** are entered into the reimbursement form and additionally combined into one supporting PDF
 - **Mileage claims** are inserted into the reimbursement form as calculated entries
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5. Human Review and Final Approval

Before anything is sent, the user receives a final review step.

Review screen should show

- all claimed rides
- all invoices found
- all receipts uploaded
- all mileage entries
- reimbursement form preview
- final email draft to SVA
- all attachments

Approval mechanism

If the user wants, they can manually review everything and then confirm the package via an **“Accept” button**.

After approval:

- the email is sent to SVA
- the full monthly package is archived in the system
- the month is marked as completed

Important governance rule

The agent must **not send anything automatically without user approval**. This fits the course logic of agentic systems with human oversight at important decision boundaries.

6. Suggested Multi-Agent Structure

To make the prototype clear and modular, the workflow can be split into several specialized agents.

1. Planning Agent

Responsible for:

- storing appointments
- structuring rides
- preparing booking requests for TixiTaxi
- updating ride schedules

2. Communication Agent

Responsible for:

- sending booking emails to TixiTaxi
- receiving confirmations
- identifying time deviations
- drafting escalation text if deviation exceeds 30 minutes

3. Invoice Verification Agent

Responsible for:

- detecting TixiTaxi invoices in email
- extracting billing data
- matching invoices against stored rides

- flagging discrepancies

4. Receipt Processing Agent

Responsible for:

- processing manually uploaded receipts
- extracting relevant data
- storing receipts in the correct category

5. Mileage Agent

Responsible for:

- calculating self-driven trip distances
- computing reimbursement amounts
- storing mileage claims

6. Reimbursement Agent

Responsible for:

- filling the SVA form
- compiling all monthly claims
- preparing the final reimbursement package

7. Review & Submission Agent

Responsible for:

- creating the final email draft
- presenting all documents for review
- sending only after user approval

This modular design is also more in line with the course recommendation to avoid one “mega-agent” and instead use several focused agents coordinated through orchestration.

7. End-to-End User Journey

In simple terms, the monthly journey looks like this:

1. The user enters future rides
2. The agent sends them to TixiTaxi
3. Confirmed rides are stored in the system
4. The user manually cancels rides by phone if needed
5. At month-end, the agent retrieves TixiTaxi invoices from email

6. The user uploads other receipts manually
7. The mileage agent adds private car trips
8. The system checks, organizes, and fills the reimbursement form
9. The agent drafts the final SVA email with attachments
10. The user reviews and clicks **Accept**
11. The claim is sent and archived